

Alan Guo

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EDUCATION

University of British Columbia

Vancouver, BC

Engineering Physics, BAsC. | GPA: 4.33/4.33

In Progress

- **Awards:** Trek Excellence Scholarship (2025, 2026); Thomas Beeching Scholarship (2026)
- **Relevant Coursework:** Introduction to Instrument Design, Signals and Systems, Electronic Circuits, Sensors and Actuators in Microsystems, Experimental Techniques, Software Construction

EXPERIENCE

Agriculture Robot Electrical Design Team Member

Jan. 2026 – Present

UBC Agrobot

Vancouver, BC

- Developed a closed-loop steering system for an autonomous agricultural robot by integrating stepper motors, position sensors, and embedded control algorithms to achieve $\pm 1^\circ$ wheel angle tracking accuracy.
- Developed startup calibration methods using magnetic encoder feedback, custom sensor PCB design, and mechanical limit switch homing to achieve repeatable zero-position initialization
- Designed actuator characterization procedures by testing stepper motors across 100–1500 RPM operating ranges and evaluating torque performance to guide steering system selection

Warehouseman

Jun. 2025 – Aug. 2025

Fasteel

Langley, BC

- Gained practical exposure to materials handling, manufacturing workflows, and industrial safety standards

PROJECTS

Autonomous Robotics Design Project

Jun. 2026 – Aug. 2026

- Reduced wiring complexity and improved system reliability by designing and assembling custom KiCad PCBs for IR sensing, metal detection, and embedded control, followed by soldering and PCB bring-up
- Developed ESP32 C++ firmware for motor, sensor, and actuator control, integrating ADC, PWM, SPI, and UART peripherals while managing a fully 20+ GPIO pins for autonomous robot operation
- Enabled autonomous object detection by developing a Python computer vision system on a Raspberry Pi Zero W that identified targets and transmitted results to the ESP32 through UART within 300 ms
- Improved system integration by designing power distribution and wiring connections across microcontrollers, sensors, motor drivers, and actuators while troubleshooting faults with multimeters and oscilloscopes

ESP32 Smart Alarm Clock & Custom PCB

Sep. 2026 – Present

- Designed a custom 2-layer embedded PCB in KiCad integrating 8+ hardware peripherals including ESP32, OLED display, RTC, sensors, buzzer driver, and USB-C power circuitry
- Transformed a product concept into a functional embedded prototype by completing requirements definition, breadboard validation, schematic capture, PCB layout, assembly, firmware development, and system testing
- Optimized component selection by evaluating datasheets to ensure PCB compatibility and design reliability
- Created embedded C/C++ firmware implementing user interfaces, alarm scheduling, menu navigation, and peripheral control for a standalone smart device

Servo Motor Circuit

Nov. 2025 – Dec. 2025

- Designed a closed-loop servo motor controller integrating analog feedback, digital logic, an 8-bit R-2R DAC, op-amp error amplification, and BJT motor drivers to regulate motor position
- Implemented proportional-integral feedback control to minimize motor error by adjusting output response based on potentiometer-defined position setpoints
- Converted digital motor feedback into analog control signals using an 8-bit counter and R-2R DAC to enable closed-loop speed regulation

TECHNICAL SKILLS

Electrical Design: KiCad, Schematic Capture, PCB Layout, PCB Bring-Up, Protoboarding, Soldering, Component Selection, Multimeters, Oscilloscopes

Embedded Systems: ESP32, I²C, SPI, UART, GPIO, PWM, Sensor and Actuator Integration

Mechanical Design: CAD, Laser Cutting, 3D Printing

Software: C++, C, Java, Python, MATLAB, Excel, Git, Linux